



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI PACHIAMMAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

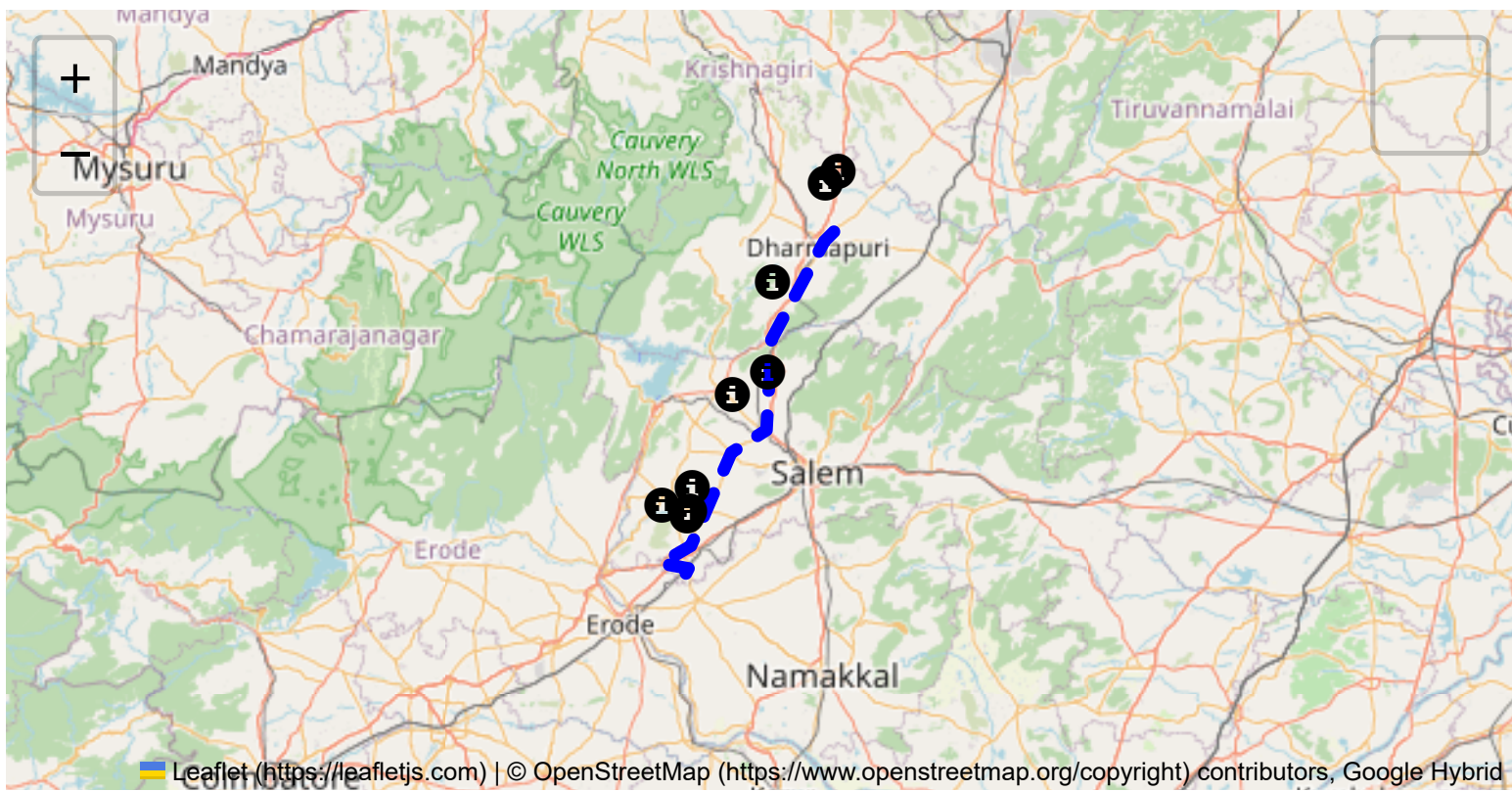
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

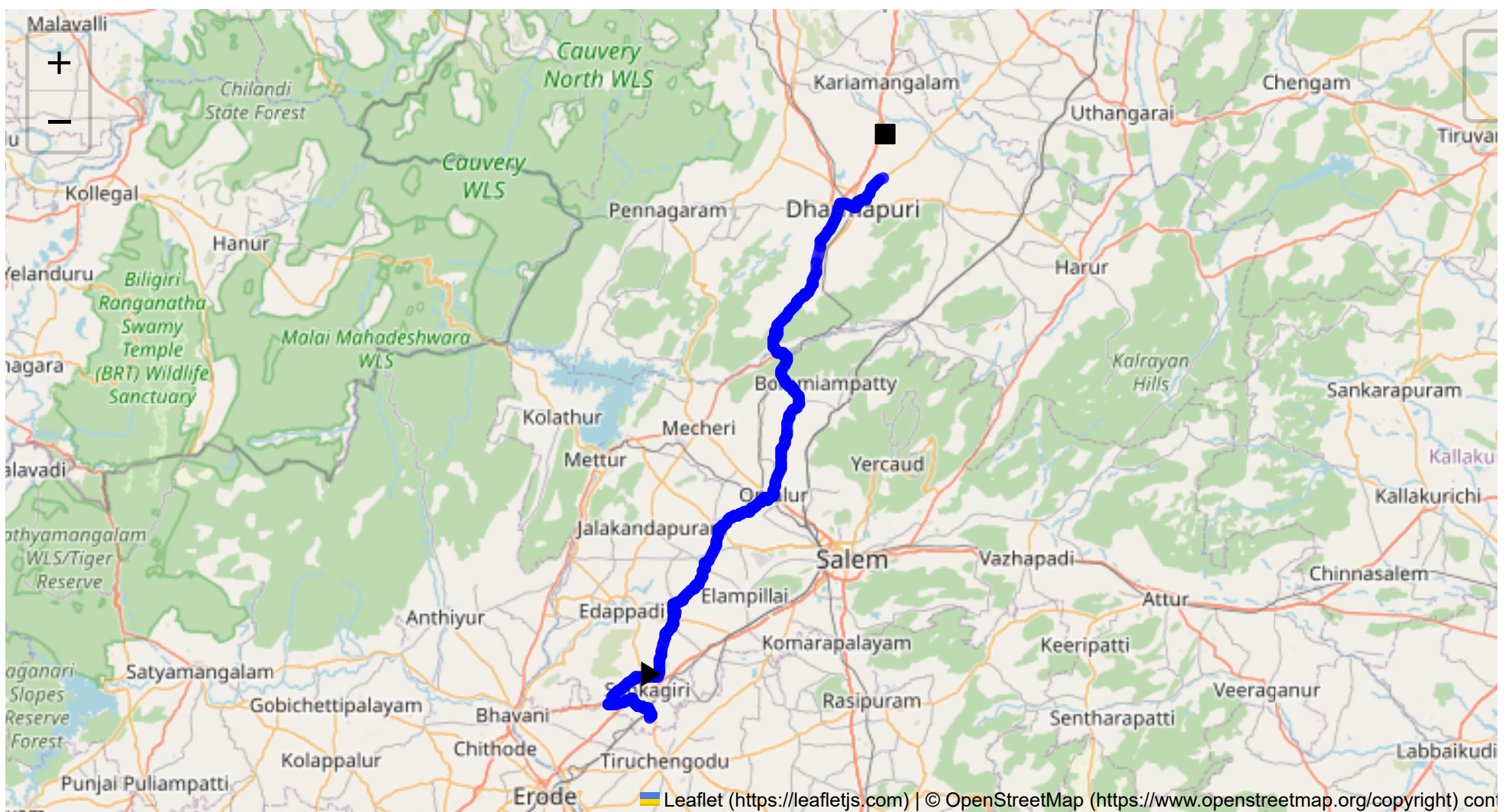
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 114.01 km
Estimated Duration: 2.5 hours
Adjusted Duration (Heavy Vehicle): 3.1 hours
Start: (11.4381, 77.8734)
End: (12.175832, 78.199905)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

The route begins in Sangagiri, Tamil Nadu, and proceeds northwards through key locations such as Salem, Omalur, Thoppur, and finally reaches Semmadakuppam in Dharmapuri. The journey covers a mix of state highways (SH) and national highways, providing a direct northward trajectory. The path intersects major roads like the Salem-Kochi Highway and SH 86, ensuring connectivity across the region.

2. Typical Weather Conditions and Potential Weather-Related Hazards:

Tamil Nadu generally experiences a tropical climate with high humidity. The region faces a hot semi-arid climate with significant temperature variations. Monsoon season (June to September) can lead to heavy rainfall, causing road flooding and reduced visibility. Cyclones or severe storms are occasional threats, especially during the monsoon.

3. Analysis of Traffic Patterns:

Traffic congestion is common in urban centers like Salem and near market areas in towns. Peak traffic typically occurs during morning and evening rush hours (8:00-10:00 AM, 5:00-8:00 PM). Roadworks or agricultural transport might add to congestion along rural segments, particularly during harvest seasons.

4. Assessment of Road Quality and Infrastructure:

Main highways (e.g., Salem-Kochi Highway) are generally well-maintained, providing smooth transit for heavy vehicles. However, regional and state roads may have uneven surfaces, potholes, and limited shoulder space, requiring cautious navigation. Rural stretches might have inadequate signage and lighting.

5. Suggestions for Alternative Routes for Emergencies:

- **Alternative Route via NH44:** For emergencies, divert to NH44; it runs parallel and offers quicker access to emergency services.
- **Local Diversions:** Through minor roads in Omalur and Thoppur that reconnect to primary highways.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

Transporting hazardous materials requires adherence to specific licenses and permissions in Tamil Nadu. Scheduled times for movement during off-peak hours are common regulations. Ensure all vehicle documentation and safety equipment comply with local laws.

7. Overview of Historical Incidents:

Occasional incidents like vehicle breakdowns or minor accidents have been recorded, particularly around Salem and Thoppur. Historical data on severe accidents involving hazardous materials is limited, suggesting a generally safe record, though vigilance is critical.

8. Environmental Considerations and Sensitive Areas:

The route cuts through agricultural areas; avoid contamination risks. In forested regions near Thoppur, look out for wildlife crossings. Ensure pollution controls to prevent hazardous discharges.

9. Analysis of Communication Coverage:

Network coverage is dependable in urban areas such as Salem. However, expect potential dead zones, especially in rural sections. Reliable communication devices and alternative connections like satellite phones are advisable.

10. Estimated Emergency Response Times:

- **Urban Segments (Salem):** 20-30 minutes.

- **Rural Areas (Thoppur, etc.):** Could extend up to 45 minutes to an hour. Closest services are in Salem and Dharmapuri.

11. Overall Summary of Risk Assessment:

The route shows a moderate risk level due to varying road conditions, potential for congestion, and occasional severe weather. A focus on preparedness for adverse scenarios, compliance with local laws, and adherence to safety protocols can mitigate most risks. Regular updates on weather and traffic situations, coupled with awareness of road conditions and emergency contacts, will further enhance safety.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.13658, 78.16171	15 KM/Hr
1	Roundabout	High	12.14589, 78.17156	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44898, 77.87410	15 KM/Hr
5	Turn	High	11.45352, 77.85698	15 KM/Hr
6	Turn	Medium	11.45847, 77.85140	30 KM/Hr
7	Turn	Medium	11.45898, 77.85135	30 KM/Hr
8	Turn	High	11.46303, 77.84945	15 KM/Hr
9	Turn	Medium	11.45543, 77.81740	30 KM/Hr
10	Turn	High	11.45631, 77.81668	15 KM/Hr
11	Turn	High	11.49383, 77.88442	15 KM/Hr
12	Turn	High	11.55608, 77.89762	15 KM/Hr
13	Turn	Medium	11.55640, 77.89913	30 KM/Hr
14	Turn	Medium	11.57677, 77.90863	30 KM/Hr
15	Turn	Medium	11.58249, 77.90829	30 KM/Hr
16	Turn	Medium	11.58294, 77.90775	30 KM/Hr
17	Turn	High	11.58320, 77.90646	15 KM/Hr
18	Turn	Medium	11.63836, 77.94819	30 KM/Hr
19	Turn	Medium	11.64176, 77.95188	30 KM/Hr
20	Turn	Medium	11.64724, 77.95118	30 KM/Hr
21	Turn	High	11.69596, 77.97122	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
22	Turn	Medium	11.71399, 77.99580	30 KM/Hr
23	Turn	Medium	11.71917, 78.01123	30 KM/Hr
24	Turn	Medium	11.73769, 78.03451	30 KM/Hr
25	Turn	Medium	11.73785, 78.03551	30 KM/Hr
26	Turn	High	11.73656, 78.03893	15 KM/Hr
27	Turn	Medium	11.74265, 78.04710	30 KM/Hr
28	Turn	Medium	11.87022, 78.08380	30 KM/Hr
29	Turn	Medium	11.87913, 78.08206	30 KM/Hr
30	Turn	Medium	11.92566, 78.06503	30 KM/Hr
31	Turn	Medium	11.93694, 78.05425	30 KM/Hr
32	Turn	Medium	11.95531, 78.04698	30 KM/Hr
33	Turn	Medium	11.95854, 78.04990	30 KM/Hr
34	Turn	Medium	11.96015, 78.04945	30 KM/Hr
35	Turn	Medium	11.96370, 78.05142	30 KM/Hr
36	Turn	Medium	11.96433, 78.05113	30 KM/Hr
37	Turn	High	11.96542, 78.04938	15 KM/Hr
38	Turn	High	11.96650, 78.04927	15 KM/Hr
39	Turn	Medium	11.96804, 78.05080	30 KM/Hr
40	Turn	High	12.14045, 78.13962	15 KM/Hr
41	Turn	High	12.14497, 78.16507	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
7	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
8	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
10	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
13	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
14	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
15	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
16	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
17	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
18	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
19	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
20	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
21	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
22	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
23	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
24	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
25	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
27	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
29	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
30	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
31	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
33	hospital	Vijaya Hospital	12.1399758, 78.1532115	30 km/h	Medium
34	clinic	Bee Tee Clinic	12.138889, 78.1576748	30 km/h	Medium
35	clinic	Sheela Clinic	12.138725, 78.1573436	30 km/h	Medium
36	clinic	Suguna Clinic	12.1386752, 78.1575468	30 km/h	Medium
37	hospital	Navina Hospital	12.1367521, 78.1610296	30 km/h	Medium
38	clinic	Bharani Neurologist	12.135969, 78.16255	30 km/h	Medium
39	hospital	Pragadeesh Hospital	12.135284, 78.1591867	30 km/h	Medium
40	hospital	D.N.V Hospital	12.1351328, 78.162869	30 km/h	Medium
41	clinic	Ragavendra Physiotherapy Clinic	12.135569, 78.1635247	30 km/h	Medium
42	hospital	Kamatchi Hospital	12.1350608, 78.1594841	30 km/h	Medium
43	clinic	Malar Siddha Clinic	12.1350485, 78.159226	30 km/h	Medium
44	hospital	Kamala Nehru Hospital	12.13404, 78.1633476	30 km/h	Medium
45	clinic	Govindaraj Clinic	12.1344914, 78.1642372	30 km/h	Medium
46	hospital	Sri Priya Hospital	12.135347, 78.164391	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
12	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
26	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
28	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
32	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.13658, 78.16171



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14589, 78.17156



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45543, 77.81740



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45631, 77.81668



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.55608, 77.89762



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58320, 77.90646



Coordinates: 11.63836, 77.94819



Coordinates: 11.64176, 77.95188



Coordinates: 11.64724, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69596, 77.97122



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71399, 77.99580



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71917, 78.01123



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73785, 78.03551



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73656, 78.03893



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74265, 78.04710



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87022, 78.08380



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87913, 78.08206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92566, 78.06503



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93694, 78.05425



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96370, 78.05142



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96433, 78.05113



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96650, 78.04927



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96804, 78.05080



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14045, 78.13962



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.14497, 78.16507
